

Using Twos, Fives, and Tens as Anchor Facts

Answer Key

Grades 2–3 Operations and Algebraic Thinking

Quick Answers

1. 14	3. 48	5. 63	7. 49
2. 60	4. 24	6. 9	8. 8

Curriculum

Level: Grades 2–3 Operations and Algebraic Thinking

3.OA.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

- 4. *If they got 14: added one more 2 instead of doubling the whole product*
 - 5. *If they got 69: took away only 1 instead of one whole group of 7*
 - 7. *If they got 42: added only one more group of 7 to the five-fact instead of two*
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1. 7 packs of 2 erasers — skip-counting by twos.

Step 1. Decide what one jump is worth before counting. Each pack is a group of 2, so every jump adds 2.

Step 2. Make 7 jumps: 2, 4, 6, 8, 10, 12, 14. Seven groups of 2 is 7×2 . 14

Why this is an anchor: counting by twos never has to be memorised — it is just every other number — so every two-fact is free.

2. 6 full racks of 10 markers — skip-counting by tens.

Step 1. Each ten-frame is one complete group of 10, so the running total goes up by a whole ten each time rather than one marker at a time.

Step 2. Count the racks: 10, 20, 30, 40, 50, 60. Six groups of 10 is 6×10 . 60

Why this is an anchor: a ten-fact is the digit with a zero after it, which is why tens are the anchor every other fact leans on.

3. From $8 \times 5 = 40$ to 8×6 .

Step 1. Compare the two facts before computing anything. Both count groups of 8; the second one has *one more group*, not one more counter.

Step 2. So $8 \times 6 = 40 + 8$, which is 48. 48

Picture check: 5 circled groups of 8 dots plus one extra circled group of 8 — the drawing shows the single step from the anchor.

4. Doubling the two-fact $6 \times 2 = 12$ to get 6×4 .

Step 1. 4 is two 2s, so six groups of 4 is the same as six groups of 2 counted twice. Doubling the product is legal for exactly that reason.

Step 2. Double the anchor: $12 + 12 = 24$.

24

Watch out: a student who adds one more 2 gets 14. That step adds a single counter, but the jump from a two-fact to a four-fact adds a whole second copy of the picture.

5. From the ten-fact $7 \times 10 = 70$ down to 7×9 .

Step 1. Ten groups is one group too many, so start high and remove a group instead of building nine groups from scratch.

Step 2. One group is worth 7, so $7 \times 9 = 70 - 7 = 63$.

63

Watch out: subtracting 1 gives 69. The 9 in 7×9 counts *groups*, so what comes off is one whole group of 7 — the difference between removing a row and removing a dot.

6. 45 beads at 5 beads per bracelet.

Step 1. Division asks how many jumps of 5 fit in 45, so count up on the five-anchor and track the jumps rather than sharing the beads out one by one.

Step 2. 5, 10, 15, 20, 25, 30, 35, 40, 45 — that is 9 jumps, so $5 \times 9 = 45$ and $45 \div 5 = 9$.

9

Fact family: $5 \times 9 = 45$, $9 \times 5 = 45$, $45 \div 5 = 9$, $45 \div 9 = 5$.

7. Splitting 7×7 into a five-fact and a two-fact.

Step 1. There is no seven-anchor, so build one: $5 + 2 = 7$ means each row of 7 splits into a part of 5 and a part of 2, and the whole array splits the same way.

Step 2. Left part: $7 \times 5 = 35$. Right part: $7 \times 2 = 14$.

Step 3. Put the parts back together: $35 + 14 = 49$.

49

Watch out: adding only one extra 7 to 35 gives 42, which is 7×6 . The split was into 5 and 2, so two groups come back, not one.

8. Challenge: 56 stickers in rows of 7.

Step 1. Start from the anchor the student already owns, $7 \times 5 = 35$, and ask how much of the pile is still unplaced.

Step 2. $56 - 35 = 21$ stickers left over after five rows.

Step 3. Count on by sevens: 42, 49, 56 — three more jumps, so three more rows.

Step 4. Altogether $5 + 3 = 8$ rows, and the division fact is $56 \div 7 = 8$.

8

Check by multiplying: $7 \times 8 = 56$, so every sticker is used and no row is short.